Lecture 20
Confidence Levels and Critical Values

Review

1 **In "we are 95% confident", what is the 95% about?**

The method: about 95% of intervals built this way capture the parameter.

2 **What is the 95% interval for p, and when does it work?**

$\hat{p} \pm 2\sqrt{(\hat{p}(1-\hat{p}) / n)}$, when $n\hat{p} > 5$ and $n(1-\hat{p}) > 5$.

3 **Where does the 2 in "± 2 SE" come from?**

95% of a normal curve lies within 1.96 SD of the mean; 2 is 1.96 rounded.

Why 95%?

Every interval on Monday was estimate ± 2 SE, which captures the parameter about 95% of the time.

A cardiologist wants to be 99% sure instead.

(a) Should the interval get wider or narrower?
(b) What in "estimate ± 2 SE" has to change?

(a) Wider: more confidence needs a bigger net
(b) The 2 becomes a bigger number, the critical value

Confidence levels

confidence level C

the share of all possible samples whose interval captures the parameter: 90%, 95%, 99%

α

the share that miss: $\alpha = 1 - C$, split as $\alpha/2$ in each tail

critical value $z_{1-\alpha/2}$

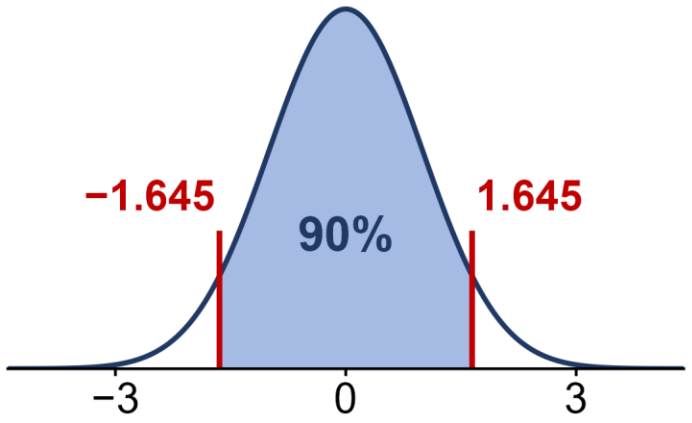
the z with area $1 - \alpha/2$ to its left; the interval is estimate $\pm z \times SE$

Where z comes from

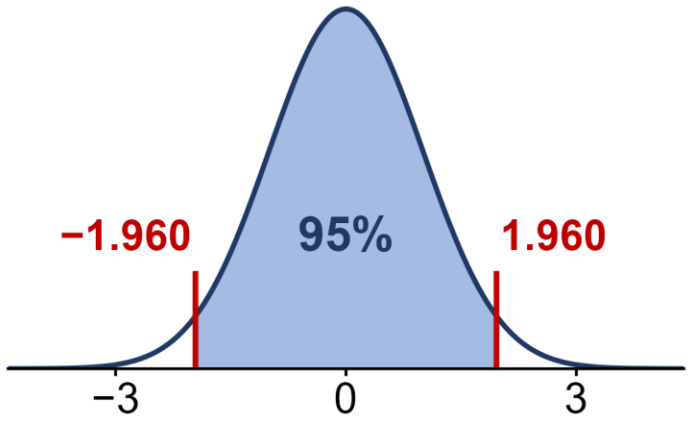
The middle C of the standard normal curve is shaded; $\alpha/2$ is left in each tail.

For a 90% interval, what area to the LEFT of z do we look up in the z-table?

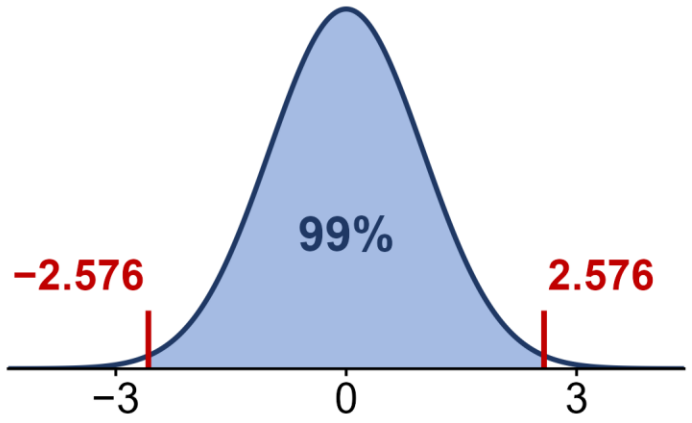
90%: 0.05 in each tail



95%: 0.025 in each tail



99%: 0.005 in each tail



90%: 1 − 0.05 = 0.95 → z = 1.645 **95%: 0.975 → 1.960** **99%: 0.995 → 2.576**

Example: critical values

Find the area $1 - \alpha/2$ inside the z-table, then read off z.

C = 80%; C = 90%; C = 99%

(a) What is $\alpha/2$ for each?
(b) What area to the left do we look up?
(c) What is the critical value?

80%: $\alpha/2 = 0.10$, area 0.90 → z = 1.28
90%: $\alpha/2 = 0.05$, area 0.95 → z = 1.645 (halfway between 1.64 and 1.65)
99%: $\alpha/2 = 0.005$, area 0.995 → z = 2.576 (between 2.57 and 2.58)

Data: a gene and a drug

Clopidogrel (Plavix), a blood thinner used after heart attacks and stents, is switched on by the liver enzyme CYP2C19. People with the CYP2C19*2 variant switch on less of it.

X = 1 if a person carries at least one copy of *2, and 0 otherwise

PERSON	POPULATION	GENOTYPE	COPIES OF *2	X
HG00403	Southern Han Chinese	G/G	0	0
HG00404	Southern Han Chinese	A/A	2	1
HG00406	Southern Han Chinese	G/A	1	1
HG00407	Southern Han Chinese	G/G	0	0
HG00409	Southern Han Chinese	G/G	0	0
…	…	…	…	…

n = 504 East Asian participants of the 1000 Genomes Project, which sequenced 2,504 people worldwide to catalog human genetic variation

Example: carriers of CYP2C19*2

Estimate p, the proportion of all individuals of East Asian descent who carry the *2 variant.

277 of n = 504 people are carriers

(a) What is $\hat{p}$? Is the check met? What is the standard error?
(b) What are the 90%, 95% and 99% confidence intervals for p?
(c) Can we say more than half of people carry *2?

(a) $\hat{p} = 277 / 504 \approx 0.550$; 277 and 227 are both > 5
SE = $\sqrt{(0.550 \times 0.450 / 504)} \approx 0.022$
(b) 90%: 0.550 ± 1.645(0.022): 0.514 to 0.586
95%: 0.550 ± 1.960(0.022): 0.507 to 0.593
99%: 0.550 ± 2.576(0.022): 0.493 to 0.607
(c) At 90% and 95%, yes; the 99% interval includes 0.5